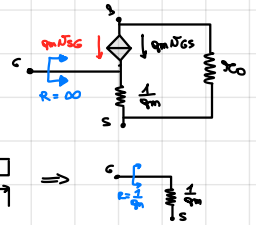
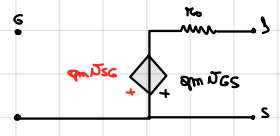
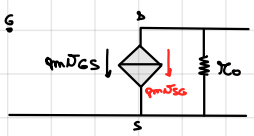


YLOS

$I_{DS} = k'_m \cdot \frac{W}{L} (V_{GS} - V_T)^2$
 $I_{DS} = k'_p \cdot \frac{W}{L} (V_{SG} - V_T)^2$
 $g_m = \sqrt{4k'_m \cdot \frac{W}{L} I_{DS}}$
 $= 2k(V_{GS} - V_T)$

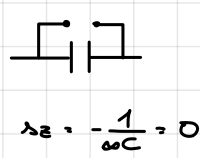
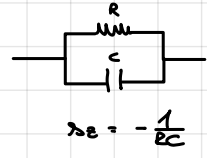
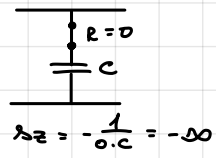
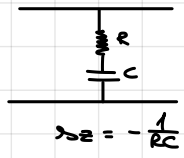
$k' = \frac{\beta'}{2}$

YLOS



ANALISI IN FREQUENZA

ZERI: $z < p \iff \lim_{s \rightarrow \infty} G(s) = 0$



POLE: mp C UNIPOLARE

METODO DEI COST. DI TEMPO

(capacite' indipendenti e interagenti)

$\sum_{i=1}^p \tau_{pi} = \sum_{i=1}^p R_i C_i \approx \tau_{PLF}$
C agente UGR. OFF

$\sum_{i=1}^p \frac{1}{\tau_{pi}} = \sum_{i=1}^p \frac{1}{R_i C_i} \approx \frac{1}{\tau_{PHF}}$
C CHARGE UGR. OFF

YILUPR: $Z_1(s) = \frac{Z(s)}{1 - A_V(s)}$

$Z_2(s) = \frac{-A_V(s) Z(s)}{A_V(s) - 1}$

TUTTI I POLI DEI GINERALS SONO ZERI DEI GROUP MA NON VICEVERSA

$|G_{GEN}|_0 = \min \{ |G_{DIRTTO}|, |G_{GRUP} G_{DIRTTO}| \}$, per $s \rightarrow \infty$ $|G_{GEN}|_{dB} \rightarrow |G_{DIRTTO}|_{dB}$

$\left| -\frac{G_{GRUP} G_{DIRTTO}}{1 - G_{GRUP}} \right| = \begin{cases} |G_{DIRTTO}| & \omega < \omega_T \\ |G_{GRUP} G_{DIRTTO}| & \omega > \omega_T \end{cases}$
 $\left| \frac{G_{DIRTTO}}{1 - G_{GRUP}} \right| = \begin{cases} 0 & \omega = \omega_T \text{ (Goup} \gg 1) \\ |G_{DIRTTO}| & \omega > \omega_T \end{cases}$

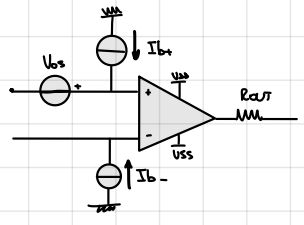
$|A_2| \omega_1^{\pm m} = |A_1| \omega_2^{\pm m}$
PUBBILICAZIONE
 $|A_2| \omega_2^m = |A_1| \omega_1^m$

GBWP = $|A(s)| \cdot \omega_L = \omega_T$

$m = +1 \rightarrow +20 \frac{dB}{dec}$

$\phi = 180^\circ - \angle G_{GRUP}(\omega_T) = 180^\circ - \arctan\left(\frac{\omega_T}{\omega_p}\right) + \arctan\left(\frac{\omega_T}{\omega_z}\right) \dots$

OPAMP

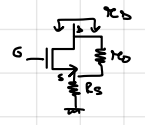


- ICMR: $V_{CM}(t) \in (V_{CMmin}, V_{CMmax})$
- Output Voltage Swing: $V_{OUT}(t) \in (V_{OUTmin}, V_{OUTmax})$
- Maximum Output Current: $I_{OUT}(t) \in (I_{OUTmin}, I_{OUTmax})$
- SR: $\left| \frac{dV_{OUT}(t)}{dt} \right| \leq SR$

$V_+ = V_{CM} + \frac{V_d}{2}$
 $V_- = V_{CM} - \frac{V_d}{2}$

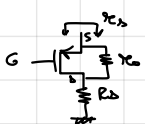
$V_{OS} = V_{OUT}|_{V_{IN}=0}$

GUARDANDO STADI EL. CON RL



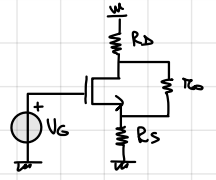
$r_{D} = (r_{o} + R_s) + g_m r_{o} R_s$

se $R_s = 0 \Rightarrow r_D = r_o$



$r_{S} = \frac{r_o + R_D}{1 + g_m r_o} \approx \frac{1}{g_m}$

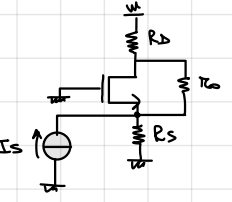
$(= \frac{1}{g_m} \text{ se } r_o \gg R_D)$



$I_{DS} = \frac{V_{GS}}{\frac{1}{g_m} (1 + \frac{R_s + R_D}{r_o})}$

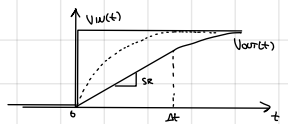
$\frac{V_d}{V_G} = \frac{-R_D}{\frac{1}{g_m} (1 + \frac{R_s + R_D}{r_o}) + R_s}$

$\frac{V_s}{V_G} = \frac{R_s}{\frac{1}{g_m} (1 + \frac{R_s + R_D}{r_o}) + R_s}$



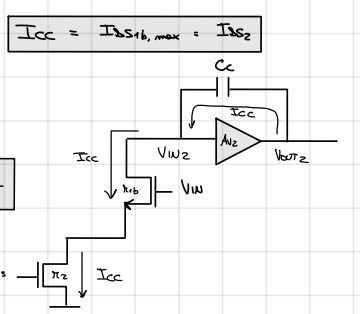
$V_o = I_s \cdot \frac{R_s R_D}{R_s + \frac{r_o + R_D}{1 + g_m r_o}}$

SR



$t < \Delta t \rightarrow$ la pendenza della retta (integrata dal quadrato) è la SR = $\frac{I_{CC}}{C_c}$
 $t > \Delta t \rightarrow$ riprende la retroazione e per $t \rightarrow +\infty$ $V_{OUT} \rightarrow V_{IN}$ (inversione)

$SR = \max_{V_{OUT}(t)} \left\{ \left| \frac{dV_{OUT}(t)}{dt} \right| \right\} \approx \frac{I_{CC, max}}{C_c} = \frac{I_{DS2}}{C_c}$



$$R_{IN} = \frac{1}{g_{m1}}$$

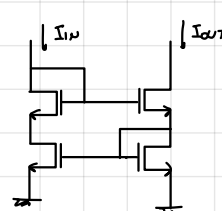
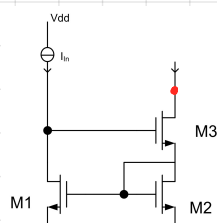
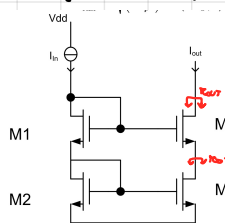
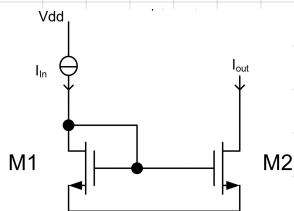
$$R_{12} = \frac{1}{q_{m1}} + \frac{1}{q_{m2}}$$

$$R_{12} = \frac{q_{m2} + q_{m3}}{q_{m2} q_{m3}}$$

$$R_{OUT} = R_{O1}$$

$$P_{out} = (K_{03} + K_{04}) + q_{m4} K_{03} K_{04}$$

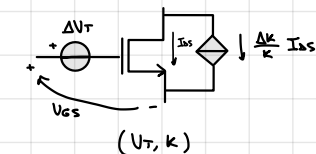
$$R_{OUT} = r_{o3}(1 + r_{o1}g_{m3})$$



SPEKTRO :

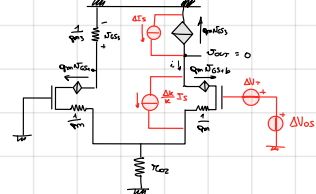
$$\frac{\sigma I_{ovr}}{I_{REF}} = \sqrt{\left(\frac{\sigma(k)}{k}\right)^2 + \left(\frac{2\sigma(V_T)}{V_{OV}}\right)^2}$$

$$\frac{\sigma(\Delta k)}{k} = \frac{\beta}{\sqrt{WL}}$$



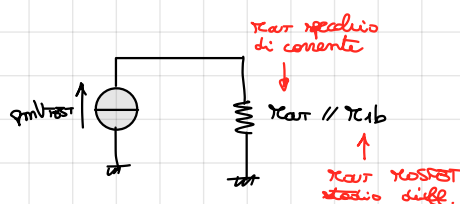
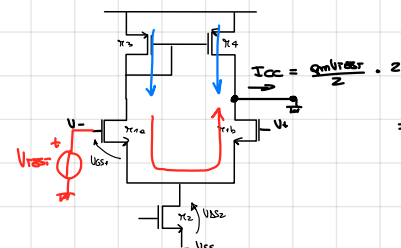
$$\sigma^2(I_{out}) = \underbrace{I_{eff}^2}_{\frac{I_{out}}{N}} \cdot \left(\frac{\sigma(u)}{K} \right)^2 + \underbrace{\left(\frac{2 I_{eff}}{V_{ov}} \right)^2}_{I_{out} = N I_{eff}} \sigma^2(V_{T1}) = q_m^2 \sigma^2(V_{T1}) + I_{eff}^2 \left(\frac{\sigma(u)}{K} \right)^2$$

N : rapporto lim. specul.



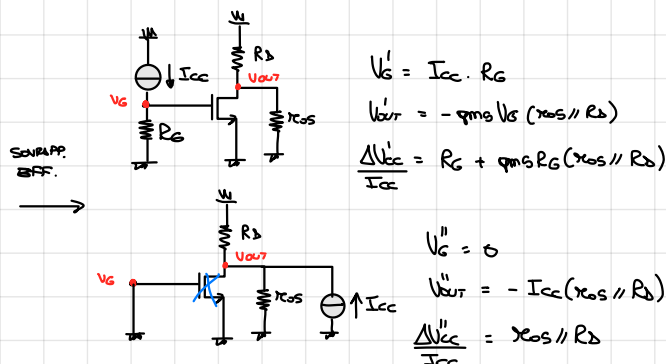
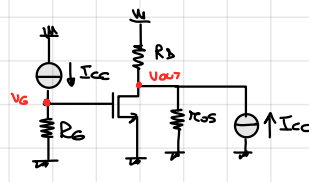
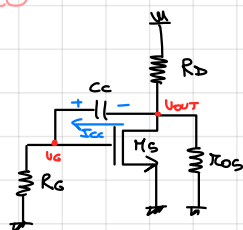
10 STADIO

EQUIVALENT LORTON



2° STADIO

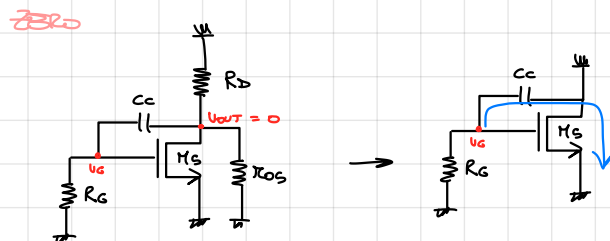
POLO



$$R_{eq} = R' + R'' = R_G + g_{m5} R_G (r_{os} \parallel R_D) + r_{os} \parallel R_D = R_G + (r_{os} \parallel R_D)(g_{m5} R_G + 1) \rightarrow \tau_{cc} = R_{eq} C_C$$

$$R_D = +10 \text{ (drain side gate)} \quad R_{eq} = (R_G + r_{GS}) + g_{mS} r_{GS} R_G$$

↑
 ruolo nel metabolismo
 della cort. di
 tempo (ne più di
 1 capacità)



$$V_G \cdot g_{Cc} = q_{ms} V_G \rightarrow g = \frac{q_{ms}}{C_c}$$

TIPS :

- tensioni ~~gibbe~~ → massa
 - nullo, monotono
- GIBBE

- Group :
 - piccolo signals
 - imp. reports
 - small parts

- Gabinetto:
- piccola stanza
 - amplificatore radio (~~2~~)
 - orologio

- poli e zeri:
- piccolo segnale
 - ingresso spento

- polarizzazione:
- ingresso speso
 - capacità aperte

$$E_{dyn} = V_{DD}^2 \cdot \alpha \cdot C_L$$

~~SSBI~~

$$C_G \equiv W L C_{ox} + 2 W C_{GS0} = S \cdot (L_{min}^2 C_{ox} + 2 L C_{GS0}) = S \cdot G_{11}$$

$$C_{UV} = S_m(1 + \epsilon) C_{x_1}$$

$$C_{GSp} = C_{Gdp} = W C_{GSO} \rightarrow C = C_{ls} + C_{Gdp}$$

CAPACITÀ DI USCITA
PASS-TRANSISTOR E INVERTER

$$C_{SB} = K_{eq} \cdot W \cdot L_{SB} \cdot C_{10}$$

$$TR: I_{DS} = \beta_m \left[(V_{GS} - V_T) V_{DS} - \frac{1}{2} V_{DS}^2 \right] (1 + \lambda V_{DS})$$

SAT: $I_{DS} = \frac{\beta_m}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

$$V_T = V_{T0} + \gamma \sqrt{2\psi_F + V_{SB}} - \sqrt{2\psi_F}$$

SE NON SPECIFICATO
VALORE 1
(è un vettore di
non indolite)

→ da calc
per m e
p nos $\Rightarrow$

→ da calc
per m e p nos